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### **Final Project Proposal: Titanic Dataset**

This proposal is conducted by Tej Govindu and Mateen Jan as part of our final project for DATA 3421: DATA MINING MGT CURATION, instructed by Dr. Qiang Zheng. We will analyze the Titanic dataset obtained from Kaggle, which provides complete passenger records of RMS Titanic, including their age, gender, ticket class fare, and their survival details. The dataset exists as a popular resource in data science because its simple structure and important features enable researchers to test their data mining techniques. The dataset will help us investigate how different elements influenced passenger survival rates and outcomes during our overall analysis.

### **Statement of the Problem**

The dataset provides detailed information about passengers aboard the RMS Titanic, which includes demographic characteristics and their survival outcomes. The central problem of this project is to determine which factors most significantly influenced passenger survival. The dataset from the disaster that happened more than a 100 years ago provides researchers with a chance to use modern data mining methods for discovering important trends. This problem is important because it allows us to explore real-world decision-making under extreme conditions while also developing predictive models that classify outcomes based on historical data.

To guide our analysis, we will focus on the following key questions:

- Which features (gender, age, passenger class, fare) had the strongest impact on survival?
- Were certain groups (women, children, or first-class passengers) more likely to survive?
- How do different variables interact with each other in influencing survival outcomes?
- Can we accurately predict whether a passenger survived using the available features?

The research questions we address will help us identify which variables have the greatest impact on survival rates while presenting the results through an understandable format. The analysis creates a dual value system because it uses historical data and statistical methods to produce valuable insights from unprocessed information. Ultimate project objectives aim to achieve two goals, which involve understanding the dataset and demonstrating how data mining techniques can solve classification problems while enabling data-based decision making.

## **Proposed Data Mining Plan**

We will solve the existing issue by implementing a data mining method that proceeds through four main stages. The first step will involve preparing the dataset so that it is clean and ready for analysis. We will address missing and inconsistent values in real-world datasets because these inconsistencies will impact our results if we do not address them. After we finish preprocessing the dataset, we will use visual methods together with statistical techniques to analyze the data and identify its hidden patterns. We will use classification models to forecast survival outcomes and assess the effectiveness of these models.

Our overall plan includes the following steps:

- **Data Cleaning and Preprocessing:**

Handle missing values (such as age), encode categorical variables ( gender and embarkation port), and remove irrelevant features that do not contribute to prediction.

- **Exploratory Data Analysis (EDA):**

Use visualizations like bar charts, histograms, and boxplots to examine trends, distributions, and relationships between variables, especially focusing on survival rates.

- **Feature Selection and Engineering:**

Identify the important variables and create new features to improve model performance.

- **Model Building:**

Apply classification techniques such as Logistic Regression and Decision Trees, and potentially Random Forest, to predict whether a passenger survived.

- **Model Evaluation:**

Assess model performance using metrics such as accuracy, confusion matrix, precision, and recall to determine which model performs best.

Our procedure will develop a dependable predictive model, which will enable us to better understand the dataset. The research design we developed will help us conduct our research into the overall problem in a systematic manner, which delivers results that answer our research questions. The project demonstrates how data mining techniques function in actual situations because it includes all steps from data preparation to the final result analysis.